THE QUOTIENT FUNCTORS ARE FPQC SHEAVES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition situation-q (canonical tag 082M). cited at statement line 2020. Standing situation.
- Lemma spaces-topologies-lemma-fpqc (canonical tag 03MR). cited at proof line 2030. Induced fpqc covering.
- Remark remark-q-sheaf (canonical tag 08IU). cited at proof line 2036. Glueing of quotients.
- Lemma spaces-morphisms-lemma-flat-permanence (canonical tag 05VY). cited at proof line 2040. Flat permanence.
- Lemma spaces-descent-lemma-finite-presentation-descends (canonical tag 060V). cited at proof line 2045. fp descent.

Source reference. Original definition 082M: sources/stacks-chow/tags.txt, source lines 10257–10257. Cited at statement line 2020: the lemma is phrased in this situation.

Source reference. Original lemma 03MR: sources/stacks-chow/tags.txt, source lines 4759–4759. Cited at proof line 2030: {X_i -> X_T} is an fpqc covering.

Source reference. Original remark 08IU: sources/stacks-chow/tags.txt, source lines 10827–10827. Cited at proof line 2036: the glued surjection F_T -> Q.

Source reference. Original lemma 05VY: sources/stacks-chow/tags.txt, source lines 7567–7567. Cited at proof line 2040: Q flat over T.

Source reference. Original lemma 060V: sources/stacks-chow/tags.txt, source lines 7739–7739. Cited at proof line 2045: Q of finite presentation.

Lemma 0.2. *In Situation 082M. The functors $Q_{\mathcal{F}/X/B}$ and $Q^{fp}_{\mathcal{F}/X/B}$ satisfy the sheaf property for the fpqc topology.*

Proof. Let $\{T_i \to T\}_{i\in I}$ be an fpqc covering of schemes over S. Set $X_i = X_{T_i} = X \times_S T_i$ and $\mathcal{F}_i = \mathcal{F}_{T_i}$. Note that $\{X_i \to X_T\}_{i\in I}$ is an fpqc covering of X_T (Topologies on Spaces, Lemma 03MR) and that $X_{T_i \times_T T_{i'}} = X_i \times_{X_T} X_{i'}$. Suppose that $\mathcal{F}_i \to \mathcal{Q}_i$ is a collection of elements of $\mathrm{Q}_{\mathcal{F}/X/B}(T_i)$ such that $\mathcal{Q}_i$ and $\mathcal{Q}_{i'}$ restrict to the same element of $\mathrm{Q}_{\mathcal{F}/X/B}(T_i \times_T T_{i'})$. By Remark 08IU we obtain

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a surjective map of quasi-coherent $\mathcal{O}_{X_T}$-modules $\mathcal{F}_T \to \mathcal{Q}$ whose restriction to X_i recovers the given quotients. By Morphisms of Spaces, Lemma 05VY we see that $\mathcal{Q}$ is flat over T. Finally, in the case of $\mathrm{Q}^{fp}_{\mathcal{F}/X/B}$, i.e., if $\mathcal{Q}_i$ are of finite presentation, then Descent on Spaces, Lemma 060V guarantees that $\mathcal{Q}$ is of finite presentation as an $\mathcal{O}_{X_T}$-module. $\square$